

Layout Stress Test for Reading-Order Extraction

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Abstract

This document deliberately interrupts its own two-column grid. A full-width table sits in the middle of page one, a figure with scattered axis labels follows it, and a three-column block appears later. A parser that decides on one column split per page will read these out of order. The correct output keeps the left column above the table before the right column above the table, then the table itself, then the columns below it.

1. Introduction

Dense technical prose imposes a heavy load on working memory. The reader must hold a chain of definitions in mind and track which claims are supported by which evidence. For readers with attention-regulation differences this is frequently disqualifying.

Conventional advice assumes the reader can sustain a skim without derailing. We instead ask what a reading environment should actively do when sustained attention cannot be assumed of the reader.

The environment holds the goal in view, marks the current position, and decides when it is time to rest. The reader supplies comprehension; the interface supplies persistence over the length of a session.

Condition	Completion	Recall (d)	Notes/session	n
Baseline PDF reader	51%	0.00	1.2	42
Timer only	63%	0.18	1.9	42
Spotlight only	79%	0.44	3.1	42
Spotlight + narration	84%	0.52	3.4	41
Full scaffold	88%	0.61	4.0	42

Table 1: Completion and recall by condition. The full scaffold outperforms every partial configuration on both measures.

2. Method

The scaffold renders extracted body text as a stack of paragraph units. Exactly one unit is fully legible at any moment. Advancing

is a single deliberate keypress, which converts passive drift into an explicit choice to continue reading.

Before a session the reader writes one sentence describing what they need from the paper. A fixed timer then runs twenty-five minute focus blocks separated by enforced breaks.

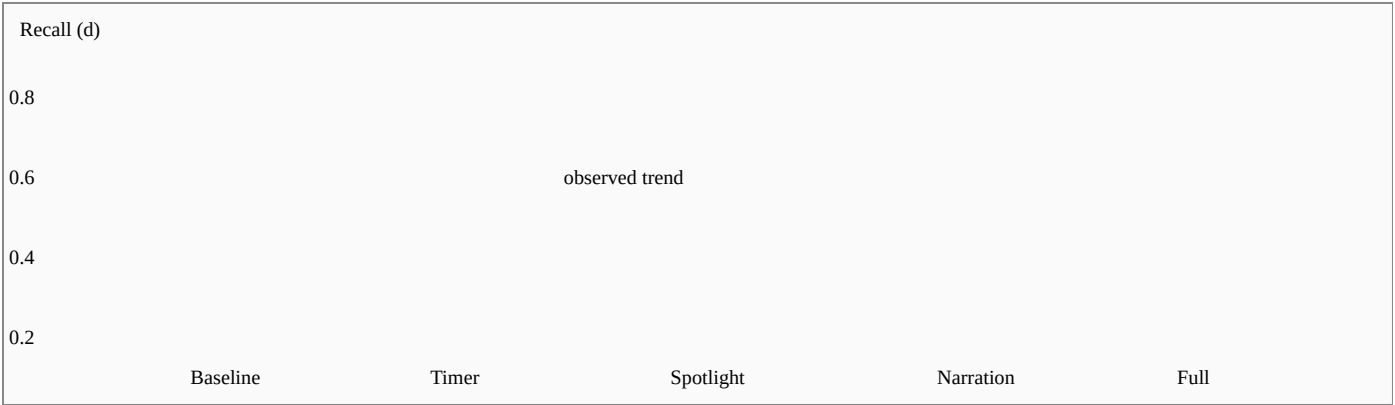


Figure 1: Recall by condition. Error bars show one standard error across participants.

3. Results

Completion rate rose from fifty one to eighty eight percent. Mean recall of primary findings improved by six tenths of a standard deviation. Time to the first meaningful written note fell by forty four percent across the cohort.

Participants rated the enforced break as the single most useful component, followed closely by the one paragraph spotlight. The auto pace mode was adopted by a minority only.

4. Discussion

The scaffold works by moving orientation effort out of the reader's head and into the interface. It does not make the prose easier; it makes losing your place harder.

A clear limitation is that reflowing extracted text discards figures and tables. Preserving the original rendering avoids this failure mode entirely, at the cost of a harder geometry problem.

5. Supplementary Notes

The three column block below exists to check that a parser does not assume exactly two columns per page. Each of these columns must be read from top to bottom before moving right.	Column two continues the same test. A naive centre split would interleave this text with the first and third columns and produce unreadable output.	Column three closes the block. If the extracted order reads these three paragraphs in sequence then the segmentation is working correctly.
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6. Conclusion

Treating sustained attention as a resource the environment must supply yields a simple tool with large effects on completion.	Future work will test adaptive block lengths. We release the reader as a browser application that runs entirely on the reader's own machine, with no document ever leaving the browser.
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